Spatial-Temporal Weapon Anomaly Detection in CCTV

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Abstract**—**Weapon detection in CCTV surveillance is challenging due to low-resolution imagery, cluttered backgrounds, and temporal motion variations. This study presents a spatial–temporal deep learning framework for detecting weapon-related anomalies in CCTV footage. Frame-level visual information is first transformed into compact spatial embeddings using a convolutional neural network. These embeddings are then organized into fixed-length temporal sequences and modeled with recurrent autoencoder architectures to learn temporal reconstruction patterns.

The proposed pipeline combines CNN-based spatial representation, recurrent sequence modeling, and autoencoder-based reconstruction for anomaly analysis. In addition to the baseline LSTM autoencoder, alternative architectures are explored to assess the effect of temporal modeling and architectural variation. Ground-truth labels are generated from XML annotations to support quantitative evaluation. The results show that spatial–temporal reconstruction provides a useful framework for analyzing weapon-related activity in surveillance video, while also revealing the limitations of reconstruction-based anomaly detection in complex CCTV scenes.

Keywords—* *CCTV surveillance, weapon detection, anomaly detection, spatial–temporal modeling, convolutional neural networks, recurrent autoencoder

I. Introduction

CCTV surveillance systems are widely used in public spaces, campuses, transportation areas, and commercial buildings to improve security and support incident monitoring. However, manually observing continuous video streams is inefficient, time-consuming, and highly dependent on human attention. For this reason, deep learning-based surveillance systems have become an important research direction for automatically detecting dangerous or suspicious activities in real time.

Weapon detection in CCTV footage is a challenging task because surveillance videos often contain low-resolution frames, distant objects, occlusions, illumination changes, motion blur, and cluttered backgrounds. In particular, weapons may appear very small compared to the full frame, making direct frame-level classification difficult. A single image may not provide enough information to distinguish between normal behavior and a potentially dangerous situation.

To address this problem, anomaly detection provides a useful alternative to direct supervised classification. Instead of requiring large amounts of labeled weapon data, an autoencoder can be trained to learn normal weapon-free surveillance patterns and then evaluate deviations through reconstruction behavior. This approach is especially useful in security scenarios, where anomalous events are rare and difficult to collect in large quantities.

Temporal modeling is also important because suspicious behavior is not always visible from a single frame. In CCTV footage, weapon-related activity may emerge gradually through movement, posture changes, or object appearance across multiple consecutive frames. Therefore, modeling frame sequences allows the system to capture temporal dependencies that are lost in isolated image-based analysis. In this project, CNN-based spatial embeddings are combined with recurrent temporal models and autoencoder-based reconstruction to analyze weapon-related anomalies in CCTV video sequences.

II. Related Work

Weapon detection in surveillance video has commonly been studied as an object detection problem, where the goal is to localize firearms in individual CCTV frames. González et al. introduced a real-time gun detection dataset collected from CCTV footage and applied a Faster R-CNN model with Feature Pyramid Network and ResNet-50 for quasi-real-time weapon detection. Their work demonstrates the importance of CCTV-specific datasets, since surveillance footage differs significantly from standard object detection images due to low resolution, camera distance, cluttered backgrounds, and small weapon appearance. More recent CCTV-oriented benchmarks also emphasize that handguns are often small, visually non-salient, and partially occluded, making weapon detection in surveillance footage a challenging computer vision task [1].

While object detection methods directly predict bounding boxes or class labels, anomaly detection approaches model normal behavior and identify deviations from learned patterns. Autoencoders are widely used in this setting because they learn compressed representations and reconstruct inputs through a bottleneck structure. Denoising autoencoders further showed that reconstruction-based learning can produce robust representations by forcing the model to recover clean inputs from corrupted versions. In video surveillance, reconstruction or prediction errors have also been used as anomaly indicators. For example, ConvLSTM-based video anomaly detection models learn temporal evolution from previous frames and derive anomaly scores from reconstruction or prediction errors. These studies motivate the use of reconstruction-based temporal modeling for detecting abnormal events in CCTV streams.

Temporal modeling is important because suspicious activity is often defined not only by object appearance but also by motion and sequence context. Recurrent neural networks address this by sharing parameters across time and maintaining hidden states that summarize previous observations. LSTM networks were introduced to reduce the vanishing-gradient problem in long temporal dependencies, making them suitable for sequence modeling tasks. GRU models provide a simpler gated recurrent alternative with fewer parameters while preserving the ability to model temporal dependencies. In this project, both LSTM- and GRU-based autoencoder variants are explored to compare how different recurrent units affect surveillance anomaly detection.

Attention mechanisms have also been widely used to improve sequence modeling by allowing models to focus on the most relevant parts of an input sequence. Bahdanau et al. introduced attention to reduce the fixed-vector bottleneck in encoder–decoder architectures by allowing soft alignment over input positions. Later, Transformer models demonstrated that attention can serve as a powerful general mechanism for sequence representation. Motivated by this, the attention-based variant in our pipeline is used to evaluate whether emphasizing important temporal segments improves reconstruction-based weapon anomaly analysis.

Compared with prior weapon detection studies that mainly rely on frame-level object detection, our work focuses on a spatial–temporal reconstruction framework. CNN-based spatial embeddings are first extracted from CCTV frames, then structured into temporal windows and processed using recurrent autoencoders. In addition to the baseline LSTM autoencoder, GRU, attention-enhanced, and dense non-temporal variants are compared as an ablation study. This allows the project to examine not only whether weapon-related activity can be detected, but also how temporal modeling and architectural choices affect reconstruction-based anomaly detection in CCTV surveillance.

III. DATASET AND PREPROCESSING

This study uses the mock [1] CCTV dataset from the real-time gun detection dataset rather than the synthetic subset. The dataset consists of surveillance footage collected from three fixed CCTV cameras placed in different locations of the same environment. These cameras cover two corridors and one entrance area, resulting in different scene layouts, lighting conditions, and background objects. The use of multiple cameras introduces realistic surveillance challenges such as distant objects, cluttered backgrounds, irregular illumination, and visually confusing objects.

The mock dataset contains three camera streams. Cam1 corresponds to a corridor scene with relatively uniform lighting and several potentially confusing objects such as doors and bins. After selecting movement-based segments, Cam1 contains 607 annotated frames. Cam7 is another corridor camera with similar lighting conditions but contains more confusing background objects, including wall-mounted objects and a fire extinguisher. This camera provides 3511 annotated frames. Cam5 is located near the entrance of a university module and contains irregular lighting conditions and conflicting visual patterns such as a dark floor carpet. This camera contributes 1031 annotated frames. In total, the mock test stream contains 5149 annotated frames.

For computational efficiency, all raw CCTV frames were converted to grayscale and resized to 64×64. This preprocessing step reduces memory usage and training time while preserving the overall object shape and spatial structure required for weapon-related analysis. Instead of training the temporal models directly on the original high-resolution image frames, each preprocessed frame is passed through a custom convolutional spatial encoder. The encoder transforms each frame into a compact 128-dimensional spatial representation. As a result, the visual data are converted into numerical feature matrices of size (3615,128) for training and (5149,128) for testing.

The training set contains 3615 weapon-free frames and is used as the normal baseline for unsupervised reconstruction learning. The test set contains the full mock CCTV stream, including both normal and weapon-visible frames. To avoid label leakage, XML annotations are not used during model training. They are used only during evaluation to construct ground-truth labels.

Temporal sequences are generated from the spatial feature matrices using a sliding-window approach. Each sequence contains 10 consecutive frames, resulting in input tensors of shape (samples,10,128). Camera boundaries are preserved during sequence generation, meaning that windows are created independently within each camera stream and no sequence is allowed to combine frames from different cameras. This prevents artificial temporal transitions between unrelated scenes. After sequence construction, the training tensor has shape (3588,10,128), while the test tensor has shape (5122,10,128).

Ground-truth labels are derived from XML annotation files, where each annotation corresponds to one video frame. The annotations are processed in chronological camera order to preserve temporal consistency with the generated feature sequences. For each frame, object categories are inspected, and frames containing weapon-related classes such as handgun, short rifle, or knife are assigned to the anomalous class; all remaining frames are treated as normal. Frame-level annotations are then converted into sequence-level labels using the same 10-frame sliding-window strategy applied during temporal sequence construction. A sequence is labeled as anomalous if at least one frame within the temporal window contains a weapon annotation. This produces a temporally aligned binary ground-truth representation for evaluating the reconstruction-based anomaly detection models.

IV. SPATIAL FEATURE EXTRACTION

After preprocessing, each CCTV frame is represented using a custom convolutional neural network designed to extract compact spatial features from grayscale 64×64 images. The purpose of this stage is to transform raw visual information into a lower-dimensional representation that preserves important shape, edge, and object-related patterns while reducing the computational cost of the later temporal models.

The CNN acts as a spatial encoder. Instead of directly passing high-resolution image frames to recurrent models, each frame is processed independently through convolutional layers that learn local visual patterns. The final representation of each frame is flattened into a 128-dimensional embedding vector. This embedding serves as a compact description of the spatial content of the frame.

As a result, the original image stream is converted into numerical feature matrices. For the test set, the extracted spatial features form a matrix of size (5149,128), where 5149 represents the total number of annotated mock CCTV frames and 128 represents the learned spatial embedding dimension. Similarly, the training set is represented as a feature matrix of size (3615,128). This transformation allows the later sequence models to operate on efficient numerical representations instead of raw image pixels.

This stage is important because it separates spatial representation learning from temporal sequence modeling. The CNN captures frame-level visual information, while the recurrent autoencoder later focuses on how these visual features evolve across time.

V. Temporal Sequence Construction

To incorporate temporal context, the extracted spatial feature matrices are converted into fixed-length sequences using a sliding-window strategy. Since each individual frame is represented by a 128-dimensional embedding, a sequence of consecutive frames can be represented as a temporal tensor. In this project, each sequence consists of 10 consecutive frames, producing input samples of shape (10,128).

The sliding-window approach moves across the ordered feature matrix and groups neighboring frames into temporal windows. This converts the frame-level representation from (N,128) into a sequence-level representation of shape (samples,10,128). After sequence construction, the training data becomes (3588,10,128), while the test data becomes (5122,10,128). This structure allows recurrent models such as LSTM and GRU to learn temporal dependencies between consecutive CCTV frames.

Camera boundaries are preserved during this process. In other words, sequences are generated independently within each camera stream, and no temporal window is allowed to combine frames from different cameras. This is necessary because different cameras correspond to different viewpoints, backgrounds, and scene layouts. Mixing the end of one camera stream with the beginning of another would create artificial temporal transitions that do not exist in the real video data.

Sequence-level labels are also generated using the same temporal-window structure. Frame-level XML annotations are first converted into normal or weapon-visible labels. Then, for each 10-frame window, the corresponding sequence is labeled as anomalous if at least one frame inside the window contains a weapon-related annotation. Otherwise, the sequence is labeled as normal. This ensures that the ground-truth labels are temporally aligned with the generated input sequences used during model evaluation.

VI. Temporal Autoencoder Architecture

The core model used in this study is a recurrent autoencoder designed to reconstruct temporal sequences of CNN-based spatial embeddings. Each input sample is represented as a sequence of 10 consecutive frame embeddings, where each frame is described by a 128-dimensional spatial feature vector. Therefore, the model receives an input tensor of shape $(10, 128)$for each temporal window.

The encoder component is based on Long Short-Term Memory (LSTM) layers. The LSTM processes the sequence in temporal order and summarizes the information from consecutive frames into a compact latent representation. This latent bottleneck forces the model to learn a compressed representation of normal temporal behavior rather than simply memorizing the input sequence. Since the training data consists of weapon-free sequences, the encoder is expected to learn temporal patterns associated with normal surveillance activity.

The decoder reconstructs the original temporal feature sequence from the latent representation. Its objective is to reproduce the input sequence as accurately as possible. The reconstructed output has the same temporal structure as the input, allowing the model to compare the original and reconstructed feature sequences at the sequence level. This sequence-to-sequence reconstruction design is more suitable for temporal anomaly detection than reconstructing individual frames independently, because it preserves the relationship between neighboring frames.

The model is trained using reconstruction loss. Specifically, Mean Squared Error (MSE) is used to measure the difference between the original input sequence and the reconstructed sequence. During inference, this reconstruction error is used as the anomaly score. The general assumption is that sequences that differ from the normal patterns learned during training will produce different reconstruction behavior. Thus, reconstruction-based scoring provides the basis for detecting weapon-related anomalies in the test data.

A. *Attention-Based Extension*

An attention-based extension is included to improve the temporal representation capacity of the recurrent autoencoder. While the baseline LSTM compresses the full sequence into a latent representation, attention allows the model to assign different importance weights to different time steps. This is useful for CCTV surveillance because weapon-related cues may not be equally visible across all frames in a temporal window.

The attention mechanism helps the model focus more strongly on informative parts of the sequence, such as frames containing sudden motion, object appearance, or posture changes. By incorporating attention, the model is expected to better capture relevant temporal patterns and reduce the limitations of relying only on the final recurrent hidden representation.

B. GRU-Based Variant

A GRU-based variant is implemented as an alternative recurrent autoencoder architecture. Gated Recurrent Units (GRUs) are similar to LSTMs but use a simpler gating mechanism with fewer parameters. This makes GRUs computationally lighter while still allowing the model to capture temporal dependencies.

The GRU variant is used to evaluate whether a simpler recurrent structure can achieve comparable reconstruction performance to the LSTM-based model. This comparison is useful because the temporal windows in this project are relatively short, containing 10 consecutive frames. Therefore, a GRU may be sufficient to model the temporal dynamics while reducing model complexity.

C. Dense Non-Temporal Baseline

A dense non-temporal baseline is also included to evaluate the importance of temporal sequence modeling. Unlike the LSTM and GRU models, this baseline does not explicitly model temporal dependencies between consecutive frames. Instead, it relies on dense layers to reconstruct the input representation without recurrent memory.

This baseline provides an ablation comparison for the temporal models. If recurrent models outperform the dense baseline, it supports the claim that temporal information contributes to weapon anomaly detection in CCTV footage. Conversely, if the dense model performs similarly, it suggests that spatial features alone may already contain strong discriminative information. Therefore, the dense baseline helps isolate the contribution of temporal modeling within the overall architecture.

VII. Anomaly Detection Strategy

The anomaly detection stage is based on reconstruction behavior. Since the temporal autoencoder is trained on weapon-free surveillance sequences, it learns to reconstruct the temporal patterns of normal CCTV activity. During testing, each input sequence is passed through the trained autoencoder, and the reconstructed sequence is compared with the original input sequence.

For each temporal window, the reconstruction error is computed using Mean Squared Error between the original and reconstructed 128-dimensional feature sequences. This error is treated as the anomaly score for the corresponding sequence. In a standard reconstruction-based anomaly detection setting, sequences that deviate from the learned normal pattern are expected to produce higher reconstruction errors.

To convert reconstruction scores into binary predictions, percentile-based thresholds are estimated from the training reconstruction-error distribution. Since the training set represents normal weapon-free behavior, its reconstruction errors provide a baseline for normal sequence reconstruction. Multiple thresholds are evaluated to analyze the sensitivity of the detector under different decision boundaries.

Ground-truth labels used for evaluation are obtained from the XML annotations. Frame-level weapon annotations are converted into sequence-level labels using the same 10-frame temporal windows used by the model. A test sequence is considered anomalous if at least one frame in the window contains a weapon-related annotation.

During evaluation, we observed that weapon-visible sequences produced lower reconstruction errors than normal surveillance sequences. Therefore, the final anomaly interpretation uses an inverted reconstruction-error direction, where lower reconstruction error is associated with weapon-visible activity. This suggests that weapon-related sequences may contain more consistent temporal patterns than normal CCTV activity, making them easier for the recurrent autoencoder to reconstruct. This behavior is treated as an empirical observation of the dataset and model, and it is further discussed as a limitation of reconstruction-based anomaly detection in complex surveillance scenes.

VIII. Experimental Results

The experimental evaluation was conducted on the temporally aligned test sequences generated from the mock CCTV dataset. Each model was evaluated using the same sequence-level ground-truth labels derived from XML annotations, ensuring that all architectural variants were compared under the same evaluation protocol. The evaluation focuses on both anomaly detection performance and the effect of different architectural components, including CNN pretraining, recurrent temporal modeling, attention, and non-temporal dense reconstruction.

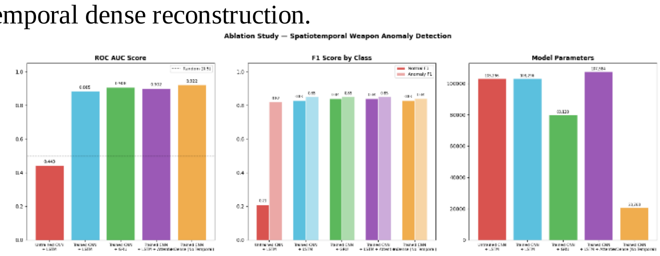

Fig. 1. *Ablation study summary comparing ROC-AUC, class-wise F1-score, and parameter count across model variants*

Table I presents the ablation study results. The untrained CNN with LSTM autoencoder performs poorly, achieving a ROC-AUC of 0.4434. This indicates that spatial feature quality has a major impact on the downstream anomaly detection task. After CNN pretraining is introduced, the LSTM autoencoder improves substantially, reaching a ROC-AUC of 0.8850 and an accuracy of 0.84. This demonstrates that meaningful spatial embeddings are essential before temporal reconstruction can be effective.

The GRU-based autoencoder achieves a ROC-AUC of 0.9044, slightly outperforming the LSTM baseline while using fewer parameters. This suggests that GRU is sufficient for modeling the short 10-frame temporal windows used in this study. The LSTM with attention also performs strongly, reaching a ROC-AUC of 0.9017, showing that attention can help the model focus on informative temporal regions.

However, its improvement over the baseline recurrent models is limited.

Interestingly, the dense non-temporal baseline achieves the highest ROC-AUC score of 0.9211, with an accuracy of 0.84. This result suggests that the CNN-extracted spatial embeddings already contain strong discriminative information for separating normal and weapon-visible sequences. Although temporal modeling remains useful for sequence-based interpretation, the dense baseline shows that spatial representation quality is one of the dominant factors in this pipeline.

Table 1. *Ablation Study Results*

Model Variant	ROC-AUC	Accuracy	Normal F1	Anomaly F1	Parameters
Untrained CNN + LSTM	0.4434	0.70	0.21	0.82	103,296
Trained CNN + LSTM	0.8850	0.84	0.83	0.85	103,296
Trained CNN + GRU	0.9044	0.84	0.84	0.85	80,128
Trained CNN + LSTM + Attention	0.9017	0.84	0.84	0.85	107,584
Trained CNN + Dense, No Temporal	0.9211	0.84	0.83	0.84	20,768

Fig. 1 summarizes the ablation study visually by comparing ROC-AUC, class-wise F1-scores, and model parameter counts. The largest performance gap appears between the untrained CNN pipeline and the trained CNN-based models, confirming that spatial pretraining is the most influential component. Among trained models, GRU, LSTM with attention, and dense reconstruction all achieve comparable classification performance.

The confusion matrices in Fig. 2–Fig. 5 provide a more detailed view of model behavior.

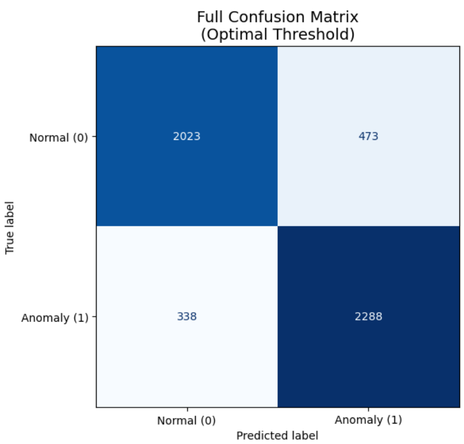

Fig. 2. *Confusion matrix of the LSTM autoencoder using the optimized anomaly threshold.*

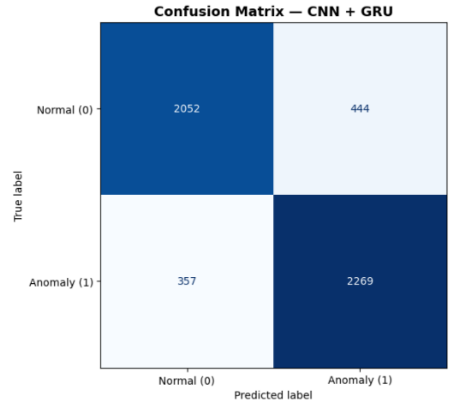

Fig. 3. *Confusion matrix of the GRU autoencoder variant*

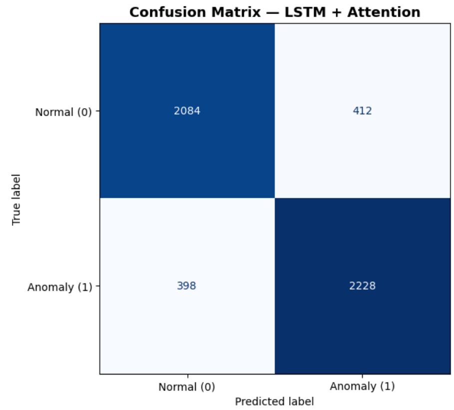

Fig. 4. *Confusion matrix of the LSTM autoencoder with attention mechanism*

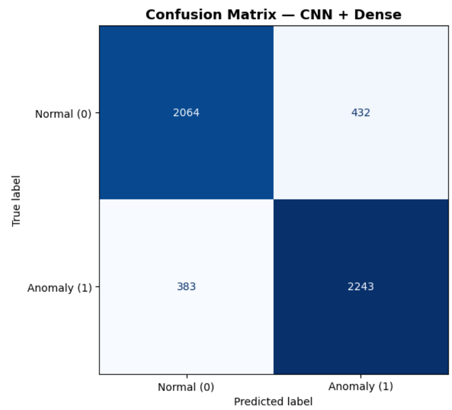

Fig. 5. *Confusion matrix of the dense non-temporal baseline*

Across the trained models, anomaly-class F1-scores remain consistently high, ranging around 0.84–0.85. This indicates that the models are effective at identifying weapon-visible sequences. The normal-class F1-scores are also strong for the trained variants, suggesting that the corrected feature extraction and label alignment pipeline produces a stable evaluation setup.

The LSTM autoencoder was further analyzed using reconstruction-based anomaly scores over time. As shown in Fig. 6, the anomaly score varies across the test sequence timeline and reflects changes in surveillance activity.

This temporal visualization is useful for interpreting how the model responds to different regions of the CCTV stream, beyond aggregate metrics alone.

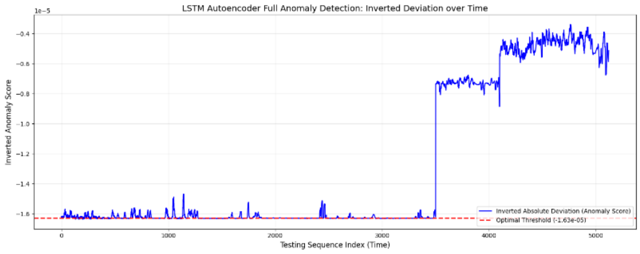

Fig. 6. *Temporal anomaly score timeline produced by the LSTM autoencoder*

Overall, the experiments show that the proposed spatial–temporal framework can effectively model weapon-related activity in CCTV footage. The ablation study highlights three main findings. First, CNN-based spatial feature extraction is essential for strong performance. Second, recurrent models such as LSTM and GRU provide reliable temporal reconstruction performance, with GRU offering a favorable balance between accuracy and parameter efficiency. Third, the dense non-temporal baseline performs surprisingly well, suggesting that the extracted spatial embeddings carry substantial discriminative information even before temporal modeling.

For the final discussion, the dense baseline obtains the highest ROC-AUC, while GRU provides the best trade-off between temporal modeling and model complexity. Therefore, the results support the use of trained CNN embeddings as the foundation of the pipeline and show that both recurrent and non-temporal reconstruction strategies are valuable for CCTV weapon anomaly detection.

IX. Discussion

The experimental results show that the quality of the spatial representation has the strongest effect on the overall anomaly detection performance. The untrained CNN with LSTM autoencoder produced weak results, while the models using trained CNN embeddings achieved substantially higher ROC-AUC and F1-scores. This indicates that meaningful frame-level spatial features are necessary before temporal reconstruction can provide useful anomaly signals. In other words, the recurrent models depend heavily on the quality of the visual embeddings they receive.

Among the temporal models, the GRU and LSTM with attention variants performed similarly to the LSTM baseline. The GRU model achieved strong performance with fewer parameters, suggesting that the 10-frame temporal windows used in this study do not require a highly complex recurrent structure. The attention-based model also performed competitively, but its improvement over the baseline was limited. This may be because the temporal windows are relatively short, reducing the need for a stronger attention mechanism.

One important observation is that the dense non-temporal baseline achieved the highest ROC-AUC among the tested variants. This result suggests that the CNN-extracted spatial embeddings already contain highly discriminative information for separating normal and weapon-visible sequences. Therefore, the contribution of temporal modeling may be more useful for interpretability and sequence-level analysis than for improving raw classification performance in this specific setup. However, this does not mean that temporal modeling is unnecessary in general; longer video sequences, more complex motion patterns, or real-time deployment scenarios may benefit more from recurrent and attention-based architectures.

The reconstruction-based anomaly detection strategy also has limitations. Autoencoders do not directly learn a weapon class; instead, they learn reconstruction behavior from the training data. As a result, reconstruction error may reflect general visual or temporal regularity rather than weapon abnormality alone. In this dataset, weapon-related activity may appear in structured segments, while normal surveillance behavior can be visually diverse. This can affect how reconstruction scores should be interpreted. Therefore, reconstruction-based anomaly detection should be combined with careful threshold calibration and ground-truth evaluation when applied to surveillance tasks.

Another limitation is that the model operates on compact 128-dimensional embeddings rather than original high-resolution frames. This improves computational efficiency but may remove fine-grained visual details that are useful for detecting small weapons. Since CCTV weapons are often small, partially occluded, or visually similar to background objects, preserving more local visual information could further improve detection performance. A future model may combine spatial embeddings with object-level localization or region-based features.

Overall, the results suggest that the proposed framework is effective as a spatial–temporal analysis pipeline for CCTV weapon anomaly detection. The strongest contribution of the project is the integration of CNN-based spatial encoding, temporal sequence construction, reconstruction-based anomaly scoring, and ablation-based architectural comparison under a consistent evaluation protocol.

X. Conclusion and Future Work

This study presented a spatial–temporal deep learning framework for weapon anomaly detection in CCTV surveillance footage. The pipeline first transforms surveillance frames into compact spatial embeddings using a CNN-based encoder. These embeddings are then organized into fixed-length temporal sequences and evaluated using reconstruction-based autoencoder models. XML annotations are used only during evaluation to construct temporally aligned ground-truth labels, while the models are trained without direct access to weapon labels.

The experiments show that trained CNN spatial embeddings significantly improve anomaly detection performance compared with untrained spatial features. Recurrent models such as LSTM and GRU provide stable sequence-level reconstruction performance, and the GRU variant achieves a favorable balance between performance and parameter efficiency. The attention-based extension also performs competitively, while the dense non-temporal baseline demonstrates that spatial embeddings alone contain strong discriminative information in this dataset.

The ablation study highlights that CNN pretraining is the most influential component in the proposed pipeline. While temporal modeling remains important for sequence-level interpretation, the results indicate that high-quality spatial representations are essential for reliable weapon anomaly detection. The final framework demonstrates the feasibility of combining CNN feature extraction, temporal sequence modeling, and autoencoder reconstruction for analyzing weapon-related activity in CCTV footage.

Future work can extend this study in several directions. First, the model can be evaluated on larger and more diverse surveillance datasets to test generalization across different camera angles, lighting conditions, and environments. Second, object localization methods can be integrated with the reconstruction framework to better detect small or partially occluded weapons. Third, longer temporal windows or multi-scale temporal models can be explored to capture more complex activity patterns. Finally, hybrid approaches that combine reconstruction-based anomaly scoring with supervised classification may improve robustness and reduce false alarms in real-world CCTV surveillance systems.

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